

IEEE WIE ILS 2026

ROUND 1 IDEA/SOLUTION DOCUMENT

National Hackathon

IDEATE. INNOVATE. IMPACT.

POWERING SOLUTIONS. EMPOWERING WOMEN. SHAPING TOMORROW.



IEEE WIE ILS 2026

National Hackathon

**Empower to Inspire: Thriving Together for an Inclusive
and Sustainable Future**

Submission Deadline: 25th August 2026

Submit as a single PDF, 12 pages max (including all sections, tables, and declaration)

Team Name HACKBROS	Track TRACK 3: SHELEADS
Problem Statement # / Title : 5. Safe Route Navigator Build an app that suggests the safest routes based on crime data, crowd density, lighting and reports	Solution Type: ☐ Hardware ☒ Software ☐ Hybrid

What's Included
<i>Do not alter this template's structure, section order, or formatting. Fill in only the marked fields/boxes. Total submission must not exceed 12 pages, including all tables, prompts, and the declaration page — submissions over 12 pages will not be evaluated. Be precise and to the point within each word-count budget.</i>
Team Information
1. Problem Understanding
2. Proposed Solution / Idea
3. Innovation & Uniqueness
4. Feasibility & Implementation Plan
5. Expected Impact
6. Technology Stack
7. Team Member Roles
Originality & Declaration

Judging Criteria (Round 1)

Criterion (Weight)	What judges look for
Problem Understanding (10%)	Clarity, evidence, relevance to inclusion/sustainability
Solution & Innovation (25%)	Originality vs. existing solutions; how directly it solves the stated problem
Feasibility & Implementation (20%)	Realistic plan, timeline, resource awareness, risk mitigation
Expected Impact (20%)	Reach, measurability, alignment with inclusion/sustainability goals
Technology & Execution (15%)	Appropriate tech/hardware choices, prototype/demo maturity
Team & Presentation (10%)	Clear roles, well-organized submission, originality declaration complete

Team Information

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Round 1 – Idea / Solution

1. Problem Understanding

- **Unsafe route selection:** Existing navigation systems mainly optimize for distance, time and traffic, without treating safety as a primary routing factor.
- **Static safety information:** Existing safety platforms often rely on historical crime data, periodic audits or crowdsourced reports, which may not reflect changing local conditions.
- **Changing risk:** Lighting, crowd density, local activity and safety conditions can vary significantly depending on the **time and location**.
- **Data limitations:** Official crime data may be delayed or incomplete, while community reports can suffer from inconsistent coverage and verification.
- **Last-mile safety gap:** In ride-hailing, safety support often ends when the vehicle reaches the destination, even though the user may still have to walk through an unfamiliar or poorly lit area.
- **Target users:** Women, students, solo travellers, families, night-time commuters and other users who want additional safety information while travelling.



- **Why it matters:** Inclusive mobility requires users to have access to **better-informed and context-aware travel choices**, rather than simply the fastest route.

2. Proposed Solution / Idea

ABHAYA — Dynamic, Risk-Aware Navigation

- **Dynamic Risk Engine:** Combines crime history, infrastructure, crowd/activity levels, recent reports, nearby safety points and time of travel to calculate comparative risk for road segments.
- **Personalized Safety Profiles:** Users can select profiles such as **Women/Solo Traveller, Family or Night Traveller**, allowing different safety factors to receive different priorities.
- **Risk-Aware Routing:** Represents roads as a weighted graph and calculates a practical route using safety, distance and travel-time costs rather than choosing only the shortest route.
- **Computer Vision Safety Audit:** A lightweight CV model can analyze available street imagery to identify indicators such as streetlights, visibility barriers and open establishments.
- **Police & Safe Points:** Displays nearby police stations and verified safety locations where reliable information is available.
- **Community Reports:** Allows users to report conditions such as broken lighting, isolated areas or recent safety concerns, while distinguishing verified information from unverified reports.
- **Time-Dependent Risk:** Risk values are adjusted according to the expected travel time so that the same road can have different risk levels at different times.

Uber Safety Integration

- **SafeDrop:** Suggests a comparatively safer nearby pickup/drop-off location when the exact destination point has higher identified risk.
- **RouteGuard:** Compares the expected ride route with observed movement and provides a discreet safety check when a significant deviation occurs.



- **WalkGuard:** After the ride ends, provides an optional short-duration safety mode for the user's final walking segment.
- **SOS Workflow:** Allows the user to activate emergency location sharing with selected contacts and, where authorized integrations exist, emergency services.

System Flow

Crime + Infrastructure + Crowd + Reports + Time



Dynamic Risk Engine



Risk-Weighted Road Graph



Personalized Route



SafeDrop → RouteGuard → WalkGuard / SOS

3. Innovation & Uniqueness

How ABHAYA differs

- **Dynamic instead of static:** Risk changes according to time, location and available evidence.
- **Multi-source risk:** Combines crime, infrastructure, activity, community reports and geographical context instead of relying on one safety indicator.
- **Personalized safety:** Different users can prioritize different safety factors.
- **Safety-aware routing:** Safety becomes part of the route cost alongside distance and travel time.
- **Computer Vision layer:** Reduces dependence on purely manual infrastructure audits by automatically identifying visual safety indicators.
- **End-to-end journey protection:** Extends beyond route selection into **pickup → ride → drop-off → final walking segment**.

Existing Solutions vs ABHAYA



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<i>Existing Solution</i>	<i>Limitation</i>	<i>ABHAYA Improvement</i>
<i>MySafetiPin / Safetipin</i>	<i>Safety audits and crowdsourced information may become outdated</i>	<i>Adds crime data, time-dependent risk and route-level optimization</i>
<i>Crime-based SafeRoute systems</i>	<i>Primarily depend on historical spatial crime data</i>	<i>Combines crime with infrastructure, activity, reports and user profiles</i>
<i>Conventional navigation apps</i>	<i>Primarily optimize time/distance</i>	<i>Adds safety as a routing objective</i>
<i>Ride-hailing systems</i>	<i>Journey safety largely focuses on the vehicle trip</i>	<i>Adds SafeDrop, RouteGuard and WalkGuard</i>

Plan & Impact

4. Feasibility & Implementation Plan

Phase 1 — Data & Map Layer

- *Collect available government crime datasets.*
- *Integrate road-network and geographical data.*
- *Store spatial data using **PostGIS**.*
- *Divide the road network into connected road segments.*

Phase 2 — Dynamic Risk Engine

- *Normalize crime, lighting, crowd, report and infrastructure indicators.*
- *Apply time-dependent weighting.*
- *Generate **Low / Moderate / High** comparative risk levels.*
- *Maintain timestamps and confidence levels for data sources.*

Phase 3 — Risk-Aware Routing



- Represent intersections as **vertices** and roads as **edges**.
- Assign dynamic costs to road segments.
- Implement **Dijkstra / A***.
- Balance safety against excessive distance and travel time.

Phase 4 — Computer Vision

- Train/use a lightweight CV model on suitable street imagery.
- Detect infrastructure indicators such as:
 - Streetlights
 - Visibility barriers
 - Open establishments
- Feed detected conditions into the risk engine.

Phase 5 — Uber Integration

- Implement **SafeDrop** using nearby location-risk analysis.
- Implement **RouteGuard** using expected vs observed route information.
- Implement **WalkGuard** as a post-ride safety workflow.
- Use simulated/API-based data where production API access is unavailable.

Phase 6 — Validation

- Test multiple routes under different times and risk conditions.
- Compare shortest-route and safety-aware-route results.
- Measure route-generation time and risk reduction.
- Conduct usability testing before wider deployment.

Key Constraints & Mitigation

- **Incomplete crime data** → show data coverage/confidence instead of assuming missing data means zero risk.
- **Outdated information** → timestamp data and periodically recalculate affected areas.
- **False community reports** → distinguish verified and unverified reports.
- **Privacy concerns** → use explicit consent and minimize location-data retention.

- **Third-party API limitations** → use simulated data for prototype demonstrations and integrate official APIs when authorized.

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5. Expected Impact

Primary Beneficiaries

- Women travelling alone
- Students and young commuters
- Solo travellers
- Families
- Night-time commuters
- Ride-hailing users
- Communities and local authorities

Expected Benefits

- **Better route decisions:** Users receive safety context in addition to travel time.
- **Improved awareness:** Users can identify changing risk conditions before and during travel.
- **Safer ride-hailing experience:** SafeDrop addresses unsafe pickup/drop-off locations.
- **Last-mile support:** WalkGuard extends safety awareness beyond the end of the vehicle journey.
- **Community participation:** User reports can highlight emerging infrastructure or safety concerns.
- **Infrastructure insights:** Repeated reports can help identify areas requiring attention.

Measurable KPIs

- **Risk reduction:** Difference in calculated risk between conventional and ABHAYA routes.
- **Travel-time overhead:** Additional time required to obtain a lower-risk route.
- **Route computation time:** Time taken to generate a recommendation.
- **CV performance:** Accuracy/precision of infrastructure detection.

- **User usability:** User ratings for clarity and usefulness.
- **SafeDrop suitability:** Percentage of recommended drop points satisfying predefined safety criteria.
- **Data freshness:** Percentage of risk inputs with valid timestamps.

Inclusion & Sustainability

- Supports **inclusive mobility** by allowing different users to choose safety priorities relevant to their circumstances.
- Can improve confidence in walking and shared mobility rather than requiring users to avoid travel because of safety uncertainty.
- A software-first architecture allows expansion to additional cities without requiring dedicated physical infrastructure.

Technology & Team

6. Technology Stack

Layer	Technologies
Frontend	React.js, Tailwind CSS
Maps	Leaflet, OpenStreetMap
Backend	Python, FastAPI
Database	PostgreSQL, PostGIS
Data Processing	Pandas, NumPy
AI/ML	Scikit-learn, CNN / Object Detection
Risk Engine	Python-based weighted risk model
Route Optimization	Graphs, Dijkstra, A*
Geospatial Processing	PostGIS, GeoJSON
Ride Integration	Uber APIs / SDKs where available
Emergency Workflow	Location APIs, consent-based live sharing
Deployment	Docker, Cloud

Hardware Not Applicable — **Software-only** prototype.



AI/ML & Open Source

- *ML: Used for risk modelling and infrastructure assessment.*
- *Computer Vision: Used to identify visual infrastructure indicators.*
- *Graph Algorithms: Used for explainable route optimization.*
- *Open-source components: Used for mapping, geospatial processing and software development to improve feasibility and reduce cost.*

User Validation

☐ Yes ☒ No

Formal user/field research has not yet been conducted. Initial validation will use scenario-based testing followed by structured user testing.

7. Team Member Roles

List only as many rows as your team size (1–6). Solo participants: complete just one row.

Member Name	Role	Responsibilities
Member 1	AI/ML LEAD	Crime-data preprocessing, dynamic risk model, risk scoring and ML evaluation.
Member 2	Computer Vision Lead	Street-level image analysis, streetlight/visibility detection and CV integration.
Member 3	DSA And Routing Lead	Road graph construction, dynamic edge weighting, Dijkstra/A* and route optimization.



Member Name	Role	Responsibilities
Member 4	Backend And Database Lead	FastAPI, PostgreSQL/PostGIS, data pipelines and backend APIs.
Member 5	Frontend And UX Lead	React interface, interactive maps, safety visualization, profiles and user experience.
Member 6	Uber and Safety Integration Lead	SafeDrop, RouteGuard, WalkGuard, SOS and emergency-contact workflow.

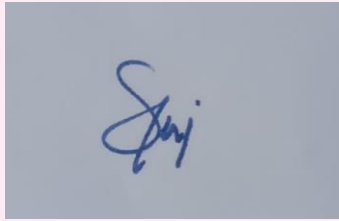
Originality & Declaration

We, the undersigned team, declare that:

- ☒ The idea/solution submitted is original and has been conceived and developed by our team.
- ☒ This submission (or a substantially similar version) has not won a prior hackathon or competition award.
- ☒ Any third-party tools, datasets, libraries, images, or references used have been duly credited within this document.
- ☒ We understand that plagiarism, fabrication of data, or misrepresentation of authorship will lead to disqualification from IEEE WIE ILS 2026.

Team Lead Name & Signature (or e-signature):

V. Suraj



Date: 24/08/2026

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